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Review Article

Influence of Footwear on Foot Comfort of Workers Engaged in Prolonged Standing Occupations: A Systematic Review

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ABSTRACT

Prolonged standing work can lead to various lower limb health issues. Footwear, as a convenient and noninvasive intervention, plays a key role in foot protection. The purpose of this review was to evaluate the impact of footwear characteristics on foot comfort among workers engaged in prolonged standing occupations. A systematic search of PubMed, Web of Science, Scopus, and the Cochrane Library was conducted until March 2025 for experimental and observational studies; the review is registered in International Prospective Register of Systematic Reviews (CRD420251003253). Given the diversity of study designs, participant characteristics, and outcome measures, results were synthesized narratively, organizing findings by footwear design factors. The synthesis indicated that soft soles reduce foot impact and fatigue but offer limited support, whereas hard soles enhance stability but may cause localized pressure and discomfort. Sole hardness should balance cushioning and support based on activity: hard soles for standing and soft soles for walking. Insole hardness must be carefully considered as semirigid arch supports can cause discomfort. Most insoles improve pressure distribution by adjusting arch loading, and some enhance intrinsic foot muscles by promoting toe engagement, reducing leg swelling. Custom orthotic insoles outperform standard ones in comfort and balance. Future research should promote interdisciplinary collaboration with footwear designers to explore key design parameters and establish a scientific framework for protective footwear.

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1. Introduction

Prolonged standing has been linked to a variety of health problems, including musculoskeletal disorders, chronic venous insufficiency, and foot deformities, with musculoskeletal disorders being the most common occupational condition, affecting approximately one-third of workers annually [1–3]. Although back, neck, and upper limb disorders have received greater attention in existing studies, lower limb health—particularly of the feet and ankles—also directly impacts work performance and quality of life [4]. However, systematic research on footwear comfort and design optimization in occupational health remains limited, which constrains the development of effective intervention strategies.

Among workers in many standing-dominant occupations, such as healthcare workers [5], manufacturing employees [6], teachers [7], chefs [8], salespeople [9], and hairdressers [10], foot pain is among the most common issues affecting work performance. About one-fifth of workers experience persistent foot pain, resulting in a substantial reduction in quality of life [11]. Most interventions for lower limb musculoskeletal disorders in occupational populations are nonsurgical [12], including footwear or sock replacement, customized orthoses, and pharmacological treatments. Nonsteroidal anti-

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inflammatory drugs can alleviate some symptoms but have limited effectiveness as not all pain originates from inflammation [13,14]; corticosteroid injections may provide short-term relief but carry risks such as fat pad atrophy and plantar fascia rupture [12].

In contrast, appropriate footwear represents a minimally invasive, long-term intervention and is therefore favored by occupational populations. However, products that effectively meet the needs of workers who stand for prolonged periods remain scarce. Surveys indicate that nurses standing for long hours spend an average of approximately 500 USD annually on shoes and compression socks to relieve pain, yet many report limited symptom improvement due to the lack of effective, targeted designs [5]. Improper footwear can lead to fatigue, pain, numbness, and even foot deformities, significantly increasing the risk of related disorders [15,16]; individuals wearing uncomfortable shoes are four times more likely to develop foot and ankle pain than those wearing comfortable shoes [17,18]. These findings underscore the critical role of occupational footwear in protecting foot health and enhancing comfort.

Although previous reviews have examined the role of cushioning materials in mitigating discomfort and fatigue from prolonged standing, footwear has often been a secondary focus. In-depth, systematic analyses targeting occupational populations are lacking, and research progress over the past five years has not been comprehensively evaluated [19,20]. Occupational footwear remains a relatively neglected area in footwear biomechanics research, limiting both theoretical advancement and the translation of findings into effective products. Accordingly, this review aims to systematically summarize relevant studies, identify key footwear design factors that alleviate discomfort and improve comfort, evaluate methodological and theoretical limitations in existing research, and propose directions for future occupational footwear design to enhance foot health among workers engaged in prolonged standing.

2. Methods

This review has been registered with the International Prospective Register of Systematic Reviews, registration number: CRD420251003253.

2.1. Search strategy

Initial scoping revealed a limited number of relevant studies; therefore, a broad time span was adopted to capture more eligible results. Studies published up to March 20, 2025, were included. The databases searched were PubMed, Web of Science, Scopus, and the Cochrane Library. The search strategy combined Medical Subject Headings with free-text terms. Since prolonged standing is often associated with work-related musculoskeletal disorders (WMSDs), this term was also included in the search. The search excluded books, conference papers, patents, and dissertations, and was restricted to studies published in English. The final search string was:

(“prolonged standing” OR “long term standing” OR “standing” OR “work-related musculoskeletal disorders” OR “WMSDs” OR “musculoskeletal disorders” OR “MSDs”) AND (“shoe*” OR “footwear*” OR “insole*”).

2.2. Eligibility criteria

As no standardized definition of prolonged standing exists, we adopted a conservative threshold of ≥40 minutes based on previous studies [2,21,22]. To focus on WMSDs in general occupational populations, studies on special jobs (e.g., military personnel, athletes) were excluded due to high-intensity or atypical workloads. Inclusion and exclusion criteria were set accordingly.

2.2.1. Inclusion criteria

Studies were included if they met the following criteria: (1) healthy participants aged 18–65 years; (2) experimental task involving standing or walking for ≥40 minutes; (3) footwear products used as an intervention; (4) outcomes included at least one subjective measure (e.g., comfort rating scale or questionnaire) or objective measure (e.g., local muscle fatigue, center of pressure, leg volume).

2.2.2. Exclusion criteria

Studies were excluded if they met any of the following criteria: (1) participants were children, adolescents, pregnant women, or elderly adults; (2) targeted special occupations requiring prolonged standing, such as military personnel or athletes; (3) intervention included confounding factors such as exercise training or foot care; (4) case studies, reviews, books, dissertations, or conference proceedings.

2.3. Quality assessment and risk of bias

The eligibility screening process was conducted in two stages. Initially, studies were screened based on titles and abstracts, followed by a full-text review according to the inclusion and exclusion criteria. Two reviewers independently conducted the screening, and any disagreements were resolved through discussion. If consensus could not be reached, a third reviewer acted as an arbitrator.

Different tools were used to assess the risk of bias based on study design. Crossover studies were assessed using the tool developed by Ding et al, which is specifically designed for crossover studies and includes nine domains, such as carryover effects [23]. Single-group studies were evaluated using the methodological index for non-randomized studies (MINORS) tool proposed by Karem Slim et al [24]. Randomized controlled trials (RCTs) were assessed using the risk of bias (RoB) tool developed by the Cochrane Collaboration, recognized as the gold standard for evaluating RCTs [25]. A brief description of each tool and the assessment results are provided in Supplementary 1.

2.4. Data synthesis

A standardized data extraction form was used to collect relevant information, including study design, participant characteristics, test setting, test duration, intervention type, and outcome measures. Due to substantial heterogeneity in participant characteristics, study duration, and methods, meta-analysis was not feasible. The included studies were categorized into three thematic groups: footwear studies, insole studies, and mixed studies. Insoles were considered separately due to their direct contact with the plantar surface, their significant impact on comfort, and the fact that, compared with shoes, they are easier to produce, replace, and control for variables, making them a common focus for researchers. Mixed studies refer to those that involved both footwear and insoles, or included footwear interventions combined with other measures. The results were then analyzed according to specific footwear design factors using a descriptive synthesis.

3. Results

A total of 3,892 articles were identified through database searches. After removing 1,664 duplicates and 175 non-original studies, 2,053 records remained. Following title and abstract

screening, 1,918 irrelevant studies were excluded, leaving 74 articles for full-text review. During full-text screening, 48 articles were excluded due to unrelated topics (39), inappropriate study design (5 acute trials, 1 with confounding factors, 1 incomplete, 1 with small sample size), or unavailable full text (1). Twenty-six studies were finally included. Some full texts were obtained by contacting authors. Two reviewers independently screened the studies with a Cohen's Kappa of 0.89, indicating good agreement. Fig. 1 illustrates the selection process, and Table 1 summarizes the key information from each study, with a more detailed version available in Supplementary 2.

3.1. Study characteristics

A total of 26 studies were included in this review, encompassing various experimental designs such as crossover studies, single-group studies, and randomized controlled trials (RCTs). Due to the lack of a standardized research paradigm in this field, some studies adopted exploratory designs. Among them, seven studies [18,26–31] contained two sub-experiments. For statistical analysis purposes, only the primary experiments closely related to the main theme of this review were retained (see Supplementary 2 for detailed analysis).

3.2. Testing methods

Researchers evaluated footwear comfort and intervention effectiveness by analyzing physiological and psychological data collected while participants wore different footwear products. In this review, the testing methods used in the 26 included studies were categorized into physiological and psychological assessments (see Table S2-1 in Supplementary Material 2).

3.2.1. Physiological testing

Physiological testing methods evolved progressively over time. Early studies recognized that discomfort was closely associated with postural control and muscle activity. Zhang et al [26] (1991) first applied electromyography (EMG), center of pressure (COP), and postural assessments. Subsequent studies introduced additional measures such as calf volume, ground reaction force (GRF), skin temperature, and cardiovascular indicators to evaluate fatigue and edema [27,33,36]. Further advancements included the integration of plantar pressure plates with motion capture systems [18], assessments of lower limb muscle strength [40,43], and measurements of interface pressures using pressure plates, insoles, or in-shoe sensor systems [46,48]. More recent approaches investigated the relationship between foot size and perceived comfort [29], step-related edema changes [49], and anaerobic metabolism through blood lactate measurement [31].

3.2.2. Psychological testing

In contrast, psychological testing methods were relatively standardized to facilitate quantitative research. Validated questionnaires, such as the Nordic Musculoskeletal Questionnaire (NMQ) [38,51] and the Borg CR-10 scale [26], were commonly employed, although they were not specifically designed for footwear comfort assessment. Therefore, some studies developed self-designed instruments, most frequently using the Likert Scale or the Visual Analog Scale (VAS) to evaluate fatigue and comfort [52]. Additional methods included ranking procedures [29] to assess footwear comfort and cognitive tests [26] to examine the

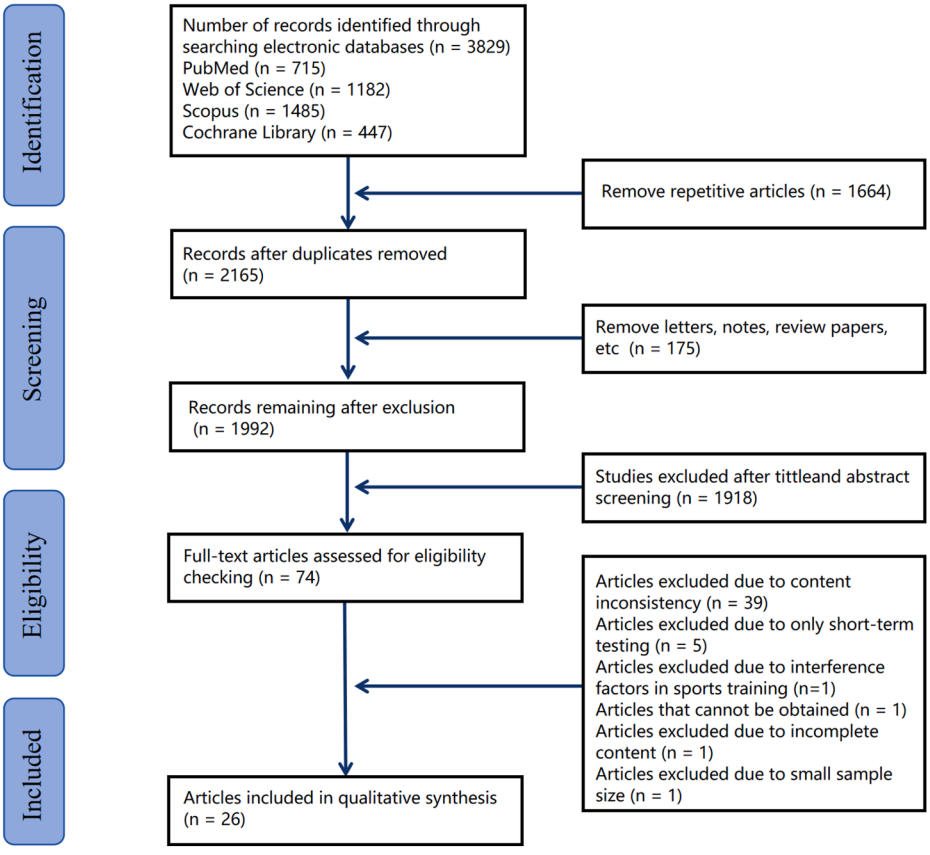


Fig. 1. Flowchart of search results and inclusion.

Table 1
Summary of key information in the article

References	Study design	Number of participants (sex)	Test site	Test duration	Types of interventions	Testing methods	
						Objective	Subjectivity
Zhang et al, 1991* [26]	CO/CO	6 (M = 5,F = 1)	S	2h	m	EMG; COP; posture evaluation	Borg; Body Part Discomfort Scale
Redfern et al, 1995 [32]	CO	14 (M = 8,F = 6)	R	1w	m		Likert Scale; Questionnaire
Hansen et al, 1998 [33]	CO	8(F = 8)	S	4h	m	EMG; calf volume (water); GRF; temperature; cardiovascular indicators	VAS
Kelaher et al, 2000* [27]	CO/SG	12 (M = 5,F = 7)	R	8w/24w	i	COP; posture evaluation; muscle strength (whole body); fatigue resistance assessment	Likert Scale
Sobel et al, 2001 [34]	SG	122 (-)	R	5w	i		Questionnaire
King, 2002 [35]	CO	22 (M = 5,F = 17)	R	1w	m		Likert Scale
Zander et al, 2004 [36]	CO	13 (M = 2,F = 11)	R	8h	m	Calf volume (Gulick)	
Orlando et al, 2004 [37]	CO	11 (M = 2,F = 9)	R	8h	m		Likert Scale
Chiu et al, 2007* [18]	CSD/CO	12(F = 12)	S	80min	m	EMG; GRF; plantar pressure; motion capture	VAS
Almeida et al, 2009 [38]	RTC	27(F = 27)	R	8w	i	Plantar pressure	NMQ
Lin et al, 2012* [28]	CO/CO	24 (M = 16,F = 8)	S/R	4h	m	COP; calf volume (Gulick)	Likert Scale
Chander et al, 2014 [39]	CO	14 (M = 14)	S	4h	s	COP	
Saadah et al, 2015 [40]	SG	16 (M = 16)	R	7h	i	Plantar pressure; muscle strength (calf)	
Almeida et al, 2016 [41]	RTC	29(F = 29)	R	5w	i	Plantar pressure	NMQ
García et al, 2016 [42]	SG	48 (M = 39,F = 9)	R	8w	i		NMQ; Likert Scale; Questionnaire
Karimi et al, 2016 [43]	CO	10 (M = 10)	S	2h	s	EMG; calf volume (Gulick); MVC	VAS
Sousa et al, 2016 [44]	RTC	30(F = 30)	R	8w	s	EMG; COP	
Karimi et al, 2017 [45]	CO	10 (M = 10)	S	2h	s	EMG; COP	
Anderson et al, 2018 [46]	CO	12 (M = 5,F = 7)	S	3h	s	EMG; calf volume (Gulick); MVC; plantar pressure; motion capture	VAS; Questionnaire
Xie X, 2018 [47]	CO	15 (M = 15)	S	40min	s	Plantar pressure	Questionnaire
Tarrade et al, 2019 [48]	SG	34 (M = 27,F = 7)	R	3w	i	COP; plantar pressure	FHSQ
Anderson et al, 2020* [29]	CO/CO	40 (-)	S/R	0/7 ~ 16h	i	Plantar pressure; foot size	Questionnaire
Nakano et al, 2020 [49]	CO	12 (M = 12)	R	8h	i	EMG; calf volume (water); MVC; step number	
Lee et al, 2022* [30]	CO/SG	58(F = 58)	S/R	0/2w	i	EMG; MVC; plantar pressure	Borg; NMQ
Chhikara et al, 2023 [50]	RTC	36(F = 36)	R	4w	i	Plantar pressure	VAS; FAAM
Fitria et al, 2023* [31]	SG/Qual	86 (M = 37,F = 49)	R	3w	s	Lactic acid	

Notes: *: The study includes two sub-experiments; -: Not described; I/II: First Experiment/Second Experiment.
CO, crossover study; COP, center of pressure; CSD, cross-sectional design; EMG, electromyographic; FAAM, Foot and Ankle Ability Measure; F, female; FHSQ, Foot Health Status Questionnaire; GRF, ground reaction force; h, hour; i, insole intervention; M, male; m, mixed intervention; min:, minute; MVC, maximal voluntary contraction; NMQ, Nordic Musculoskeletal Questionnaire; Qual, qualitative research; R, real work environment; RCT, randomized controlled trial; s, shoe intervention; S, simulated work environment; SG, single-group; w, week; VAS, Visual Analog Scale.

relationship between attention and fatigue, providing a more comprehensive evaluation of psychological responses.

3.3. Research trends

As shown in Fig. 2, although all studies involved footwear products as the primary intervention, early research often compared footwear interventions in combination with other interventions (e.g., floor materials). Insoles were considered separately because they directly contact the plantar surface, significantly affect comfort, and are easier to produce, replace, and control than shoes, making them a common focus in research. Based on this, interventions can be further categorized into footwear, insole, and combined interventions: footwear interventions refer to shoe design-related modifications excluding insoles (including footbeds), insole interventions refer to inserted components acting directly on the plantar surface, and combined interventions involve both shoes and insoles or additional interventions.

Due to the limited number of relevant studies, the search covered a broad time span (1991–2023). Early research primarily focused on combined interventions involving footwear and mats or flooring, emphasizing the impact of different intervention factors on foot health [25]. Over time, the number of publications gradually increased. As the research evolved, scholars began to focus more specifically on the effects of design features of individual intervention types on foot comfort. In terms of intervention types, insole interventions accounted for 44% of the studies, combined interventions for 30%, and shoe interventions for 26%.

Based on the affiliations of the first authors, the 26 included studies were distributed as follows: the United States (8 studies), the United Kingdom (2), China (2), Taiwan (2), Indonesia (2), Brazil

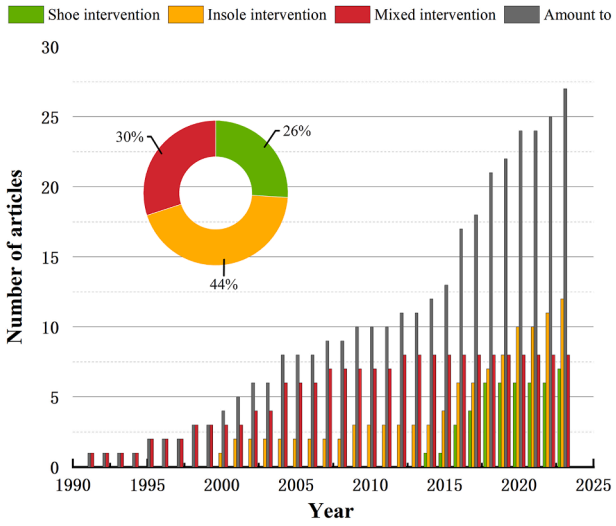


Fig. 2. The number of articles in different years.

(2), Iran (2), and one each from France, Japan, Denmark, Spain, Portugal, and India. Institutions with more than one publication included the University of Wisconsin–Milwaukee (3 studies), Faculdade de Medicina de São José do Rio Preto (2), and the University of Salford (2).

3.4. Participants

Among the 26 included studies, 7 [18,26–31] involved two experiments, but most did not explicitly state whether participants overlapped. In three studies [18,26,27], the number of participants was not reported separately for the two trials (30 if overlapping and 60 if not) and thus we conservatively treated them as overlapping and adopted the minimum value. In four other studies [28–31], participant numbers were reported separately (148 if overlapping and 208 if not), and thus we treated them as nonoverlapping and adopted the maximum value. Based on this approach, a total of 721 participants were involved in the studies (out of 852 recruited). Of these, 8 were laboratory-based studies with smaller samples (87 participants), 15 were conducted in real occupational settings with larger samples (512 participants), and 3 included both laboratory and workplace environments, involving 122 participants.

Two studies [29,34] did not report gender information. In the remaining 24 studies, 228 male and 331 female participants were included, with a higher proportion of females. This may be due to the higher prevalence of musculoskeletal problems among women in occupations involving prolonged standing [53]. Participants in laboratory-based studies were mostly aged 20–35 years and primarily university students, reflecting recruitment limitations; in occupational settings, participants ranged from 20–61 years old, and their jobs inherently required prolonged standing, making them more representative of actual occupational populations.

3.5. Task settings

Variations in task settings across studies were closely related to whether the trial was conducted in a laboratory or a real workplace. Based on the testing environment, long-standing trials were categorized into two groups: "laboratory-based trials" and "workplace-based trials." Workplace-based trials generally had longer testing durations, ranging from one working day (7 hours) to 24 weeks [33,40]. Due to the diverse occupations of participants, the frequency and proportion of standing, walking, and other tasks varied significantly across studies, making direct comparison difficult. Therefore, this section focuses only on the task settings in laboratory-based trials.

As shown in Table 2, a total of nine studies were conducted in laboratory settings, with task durations equal to or exceeding 40 minutes [21]. In these trials, the testing time was relatively short, ranging from 40 minutes to 4 hours [39,47]. Given the requirement for continuous participant involvement, which incurs a high time cost, most experiments were completed within a single day. Testing conditions included standing, walking, sitting, and resting. Overall, in comparison with prolonged static standing tasks, mixed or alternating activities are likely to reduce the load and sustained contraction time of individual muscle fibers, thereby potentially delaying the onset of fatigue [21].

3.6. Risk of bias

Risk of bias assessment revealed that seven studies included repeated experiments, of which three were excluded from evaluation due to limited relevance to the main objective of this review (e.g., methodological validation, observational or qualitative studies) [18,26,31].

Overall, after assessment with appropriate tools, the studies included in this review were generally of acceptable methodological quality, although several risks of bias were noted.

Most crossover studies considered carryover effects [54], with washout periods ranging from one day to one week, though their effectiveness remains uncertain. Randomization and allocation concealment were often insufficient, and blinding was generally not implemented, making subjective outcomes prone to bias. Some studies attempted to mitigate subjective bias through repeated measurements [32,35].

Table 2
Task setting for laboratory experiments

References	Test time (Task type)	Task work
Zhang et al, 1991* [26]	2h = 15min (Standing) × 8	Computer tasks
Hansen et al, 1998 [33]	2h (Standing)/2h=(20s (standing)+10(walking) × 240	Sorting tasks/handling tasks
Chiu et al, 2007* [18]	80min=[15min (Walking)+15min (standing)+10min (sitting)] × 2	—
Lin et al, 2012* (I) [28]	4h=[50min (Standing)+10min (sitting)] × 4	Computer tasks
Chander et al, 2014 [39]	4h = walking (allowing brief pauses of 1–2 minutes)	—
Karimi et al, 2016 [43]	2h (Standing)	Sorting tasks + jigsaw tasks
Karimi et al, 2017 [45]	2h (Standing)	Sorting tasks + jigsaw tasks
Anderson et al, 2018 [46]	3h (Standing)	Desktop tasks
Xie X, 2018 [47]	40min (Standing)	—

Notes: *: The study includes two sub-experiments; (I): First test; (II): Second test; -: Not described.
h, hour; min, minute; w, week.

RCTs were assessed using the Cochrane RoB 1.0 tool [55]. The main sources of bias stemmed from unclear randomization procedures and the lack of blinding, both of which could compromise the objectivity of the findings.

Single-group studies were assessed using the MINORS tool. The primary sources of bias included: lack of prospective sample size estimation; absence of consecutive participant enrollment—although this was considered low risk due to the specificity of the target populations; lack of blinded assessment for subjective outcomes; and inadequate follow-up duration or high attrition rates.

4. Discussion

Early studies primarily focused on comparing footwear interventions with flooring interventions. The results indicated that footwear interventions generally yielded superior effects compared to flooring, particularly in alleviating foot and knee discomfort [46], with more pronounced outcomes in these areas [34,35]. In contrast, floor materials (e.g., mats) demonstrated limited efficacy in fatigue mitigation, with benefits mainly observed in relieving lower back discomfort [26]. Investigation of the underlying mechanisms suggests that the advantage of footwear interventions lies in their direct action on the plantar surface rather than merely providing indirect cushioning and in their ability to more effectively enhance venous pump function and circulation through adjustments in gait and subtle foot muscle activity [32,33]. Combined with their greater mobility and applicability, recent research has increasingly focused on optimizing and further exploring footwear design.

Studies on footwear have found that soft-soled shoes (e.g., those made of ethylene-vinyl acetate) can significantly reduce plantar impact forces and perceived fatigue [26,33,47]. However, excessive softness may lead to insufficient arch support, increasing muscular fatigue [47]. Hard-soled footwear (e.g., work boots) offers better ankle support and balance stability [39] but may also cause localized pressure and foot discomfort [18,47]. For standing tasks, cushioning demand is low, and harder soles are more suitable for limiting plantar pressure. For walking tasks, increased cushioning is more appropriate [46]. High-top shoes provide superior ankle support and can reduce the decline in balance ability compared to low-top shoes [39]. Additionally, some studies have investigated the intervention effects of specialized footwear designs. Unstable shoes may enhance calf muscle activity and reduce lower limb swelling; however, their efficacy in improving subjective fatigue remains to be further validated [44]. Accordingly, for workers who stand for prolonged periods, the most straightforward footwear selection strategy is based on sole hardness to balance cushioning and support: hard-soled shoes help maintain plantar stability for predominantly standing tasks, whereas soft-soled shoes provide greater cushioning for mainly walking tasks.

Studies on insoles have found that they enhance foot comfort by altering midfoot loading and reducing the burden on the metatarsals and heel [29,40]. Short-term use of arch-supporting insoles (e.g., for 7 hours) can alleviate foot pain [30,40]. However, not all arch-supporting insoles yield positive effects. For instance, semirigid archsupport insoles may relieve lower back discomfort but show no significant benefit in postural control and may even increase foot discomfort [27]. Some specially designed insoles effectively alleviate foot discomfort. The Insole Padding System, an orthotic system with pads of specific shapes on the underside, can help reduce plantar pressure in the short term, particularly in the midfoot and heel regions [30]. Metatarsal pads (a type of insole with toe grips) may reduce leg swelling by promoting toe engagement during standing or walking [49]. With the advancement of 3D printing, researchers increasingly focus on personalized solutions as customized corrective insoles offer superior benefits over conventional insoles in optimizing plantar pressure distribution, enhancing balance, and improving comfort [48,50]. Different insole designs provide distinct benefits: low-hardness arch support insoles relieve plantar pressure, metatarsal pads enhance muscle activity, and customized insoles improve pressure distribution, balance, and comfort. Overall, for workers exposed to prolonged standing, most types of insoles appear to provide some degree of fatigue relief when individual arch characteristics are taken into account; however, the use of harder insoles should be approached with caution.

Although all studies included in this review focused on the effects of footwear interventions on foot comfort in occupations involving prolonged standing, the absence of a standardized research methodology in this field has led to substantial heterogeneity across studies in terms of participant characteristics, task design, and measurement indicators, thereby limiting comparability of findings.

Participant characteristics: The studies covered a wide range of occupations, including workers, police officers, students, nurses, chefs, and hairdressers. These professions differ significantly in task demands and physical conditions, resulting in varied responses. Some studies included healthy participants [39,45], while others recruited individuals with preexisting musculoskeletal disorders [18,50], with inconsistent inclusion criteria. Most studies applied only basic eligibility criteria and few adopted refined screening procedures [48], which compromises the representativeness and applicability of the findings [56].

Task design: Field studies conducted in real workplace settings better reflect actual conditions but task types are diverse and subject to daily fluctuations, reducing repeatability. Their strength lies in longer test durations, which help assess the sustainability of intervention effects. In contrast, laboratory studies offer better control over variables but typically involve shorter test periods and lack standardized protocols. To enhance representativeness, some researchers attempted to define task parameters based on real job observations. For instance, Chiu et al [18] analyzed actual nursing duties to inform experimental design. However, standardized protocols for critical factors such as standing duration and task-switching frequency are still lacking, hindering study comparability and reproducibility.

Measurement indicators: There is no consensus on how long a person must stand before discomfort occurs [21,22,57], and the timing of psychological versus physiological responses differs. Psychological discomfort tends to appear early, with some studies reporting symptoms as soon as 15 minutes, and marked increases within the first 30 minutes [26,46]. Physiological responses typically lag, with EMG changes often requiring prolonged muscle loading and becoming significant after two hours of standing [33]. Walking increases muscle activity and reduces the load and duration on individual muscle fibers [42], while static standing alone can lead to noticeable changes within 30 minutes [46]. Calf circumference is sensitive to workload during the first two hours but tends to plateau thereafter [28]. Plantar pressure often shows significant increases after 60 minutes, particularly in the midfoot region [46]. Indicators such as temperature exhibit some variation but show low sensitivity to footwear conditions and relatively weak trends [33].

Psychological assessments reflect subjective experience, are sensitive to change, but carry a higher risk of bias [58]; one study reported that 65% of participants changed their insole preference after a single day [29]. Nevertheless, psychological indicators may serve as precursors to physiological changes [59]. Physiological assessments provide objective data that are independently

valuable and irreplaceable. These two types of indicators address different aspects and should not be regarded as directly comparable. From a research perspective, it may be valuable to first delineate the footwear design parameter ranges associated with maintaining physiological comfort (objective outcomes). Within these ranges, future studies could explore how subjective comfort can be optimized. Such an approach may help ensure that footwear design not only prevents physiological impairments but also meets psychological comfort needs, thereby reducing potential conflict between subjective and objective findings.

Current research predominantly focuses on prevalence and risk factors, with a clear lack of systematic investigations into footwear interventions. This review includes only 26 studies, and the high degree of heterogeneity prevents direct comparisons. The lack of foundational research has become a major bottleneck in advancing this field. It is recommended that a standardized experimental framework be led by authoritative organizations, with a focus on clearly defining task design, including standing/walking duration, alternation frequency, specific task types, and other relevant aspects, for example, including a minimum standing duration of approximately three hours, a walking-to-standing ratio of about 2:1 for mixed tasks, and the use of well-recognized, validated comfort assessment scales specifically designed for footwear products. On this basis, the effectiveness of measurement methods can be reliably evaluated, and comparability across different studies can be ensured, providing a solid foundation for occupational foot health. Meanwhile, interdisciplinary collaboration should be promoted, inviting footwear design experts to investigate key parameters (e.g., upper materials, insoles, soles, hardness, and heel height) to develop professional protective footwear suitable for prolonged standing that effectively protects foot health.

Limitations

This review included all relevant studies published before March 2025 to maximize coverage, with two researchers participating in bias risk assessment. Nevertheless, certain limitations remain. First, due to the scarcity of research in this area, inclusion criteria were broadened as much as possible, resulting in high heterogeneity among included studies and precluding meta-analysis, thus failing to yield very definitive conclusions. Second, there remains the possibility of missed studies as the search was limited to English language publications without supplementary indexing, potentially excluding relevant research. Finally, differences across studies in participant characteristics, task design, measurement indicators, footwear interventions, and experimental settings may affect the comparability and validity of findings, further limiting their generalizability and reproducibility.

5. Conclusion

Footwear interventions can significantly alleviate fatigue and discomfort in individuals who stand for prolonged periods, while improving foot comfort. Soft-soled shoes reduce plantar impact, whereas hard-soled shoes enhance ankle stability. Insoles can improve comfort by redistributing arch load or promoting toe activity, with custom corrective insoles showing particular effectiveness in optimizing plantar pressure distribution and balance. Based on these findings, it is recommended that occupational health practices establish guidelines for footwear selection and workplace management for long-standing workers. Shoe sole stiffness should be balanced according to task and usage context, with hard soles preferred for prolonged standing and soft soles for extended walking. Most insoles designed for prolonged standing effectively enhance comfort, and both managers and users should deliberately select appropriate insoles to protect foot health, while promoting evidence-based footwear interventions in the workplace to reduce fatigue and improve foot well-being.

Current studies exhibit high heterogeneity, limiting comparability and generalizability. Future research should develop standardized experimental protocols, particularly to unify task protocols and effective comfort assessment methods. Furthermore, interdisciplinary collaboration integrating biomechanics, ergonomics, footwear design, and occupational health will facilitate the development of evidence-based protective footwear, providing scientific foot protection for long-standing workers and advancing occupational health practice and innovation.

CRediT authorship contribution statement

Lu-ping Kang: Writing – review & editing, Writing – original draft. **Tai-sheng Gong:** Writing – review & editing, Supervision. **Xiao-hong Quan:** Data curation. **Chen Zhang:** Validation, Investigation.

Data availability

The dataset used in this paper are available from the corresponding author upon request.

Statement on the use of AI tools

During the preparation of this work, the authors used generative AI tools (ChatGPT by OpenAI) for language polishing purposes. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Conflicts of interest

The authors declared that they have no conflicts of interest regarding this work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.shaw.2025.10.003.

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